

```

t_test_summary <- function(x, y) {
  # Plot density
  plot(density(x), col = 'green', main = "Density Plot of x and y")
  lines(density(y), col = 'blue')
  legend("topright", legend = c("x", "y"), col = c("green", "blue"), lty = 1)

  # Perform Welch Two Sample t-test
  ttest <- t.test(x, y)

  # Print t-test summary
  cat("Welch Two Sample t-test\n\n")
  cat("data: x and y\n")
  cat(sprintf("t = %.5f, df = %.3f, p-value = %.4f\n",
              ttest$statistic, ttest$parameter, ttest$p.value))
  cat("alternative hypothesis: true difference in means is not equal to 0\n")
  cat(sprintf("95 percent confidence interval: [%.6f, %.6f]\n",
              ttest$conf.int[1], ttest$conf.int[2]))
  cat("sample estimates:\n")
  cat(sprintf("mean of x = %.6f\n", ttest$estimate[1]))
  cat(sprintf("mean of y = %.6f\n", ttest$estimate[2]))

  # Interpretation
  if (ttest$p.value > 0.05) {
    cat("\n# Groups x and y are NOT significantly different (p > 0.05)\n")
  } else {
    cat("\n# Groups x and y ARE significantly different (p <= 0.05)\n")
  }
}

```